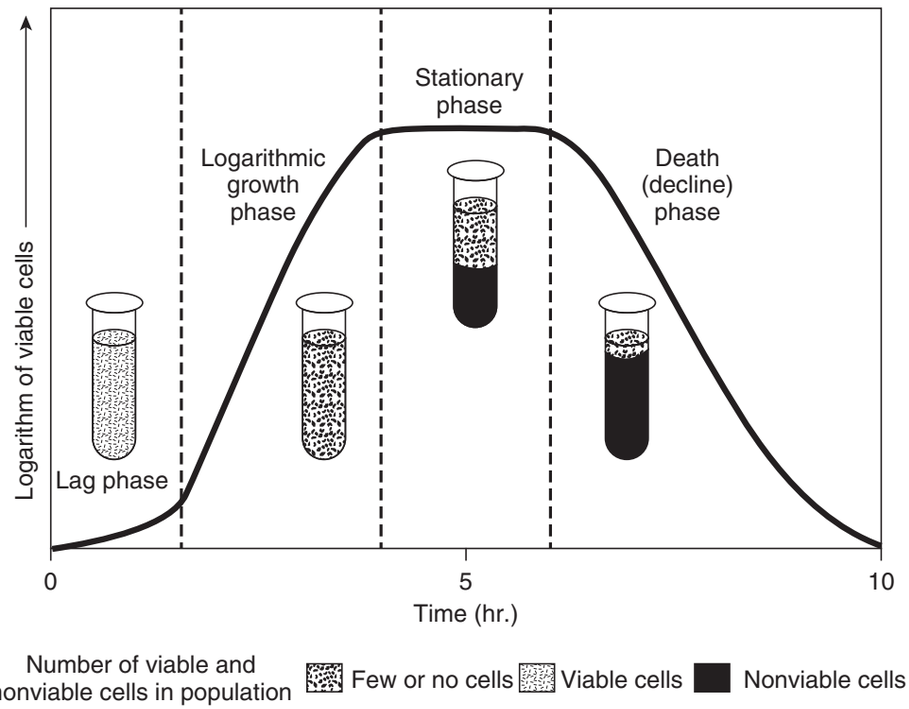


Figure 6.4: Bacterial growth curve showing phases of growth in laboratory conditions

3. Stationary Phase

As the nutrients get depleted, the cell growth stops and the growth curve become horizontal. The total number of viable cells remains constant which is due to a balance between cell division and cell death.

4. Death Phase

Nutrient deprivation and build up of wastes lead to the decline in cell numbers. The microbial population dies rapidly and logarithmically and the growth curve also stops down.

Batch culture

It is the growth of microorganisms in a fixed volume of culture medium in which nutrient supply is not renewed and wastes are not removed. It is a closed system. This can be used to study the various growth phases of microorganisms.

Continuous culture

A continuous culture is an open system with constant volume to which fresh medium is added and utilized (spent) medium are removed continuously at a constant rate. A microbial culture remains in exponential state for longer periods, for days and even weeks. This enables the researcher to learn about the physiological processes and enzymatic activities of organisms.

There are two ways by which continuous culture is operated.

a) Chemostat

b) Turbidostat

Chemostat

The chemostat operates so that the sterile nutrient medium enters the culture vessel at the same rate as the spent medium is removed. The chemostat can control growth rate and cell density simultaneously and independently of each other. Two factors play an important role in achieving this dilution rate and concentration of the limiting nutrient (a carbon or a nitrogen source like sugars or

TN_GOVT_XI_Micro_Biology_CH06.indd 77 16-02-2019 9.58.10 AM